



Stable Diffusion

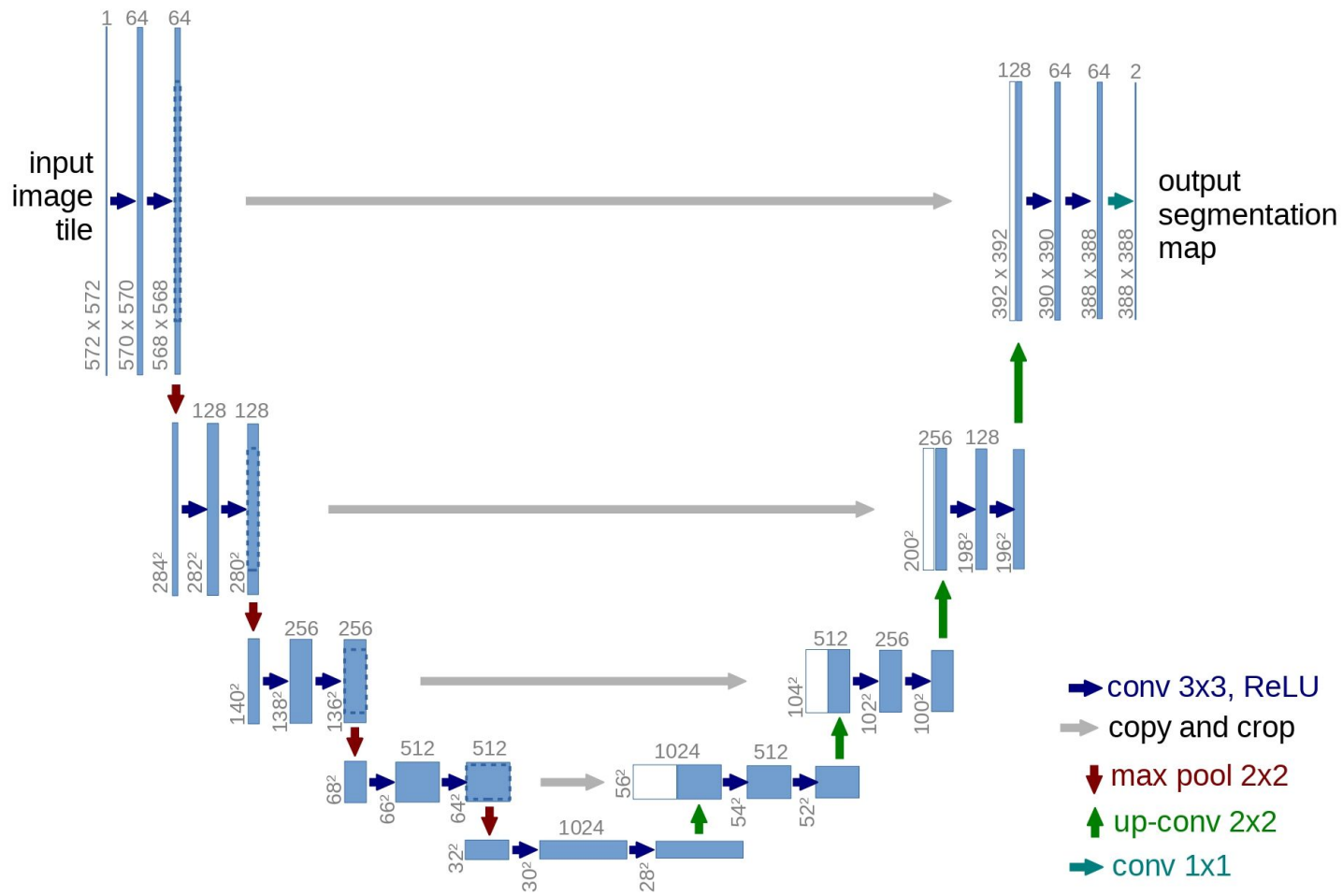


Stable Diffusion 개요

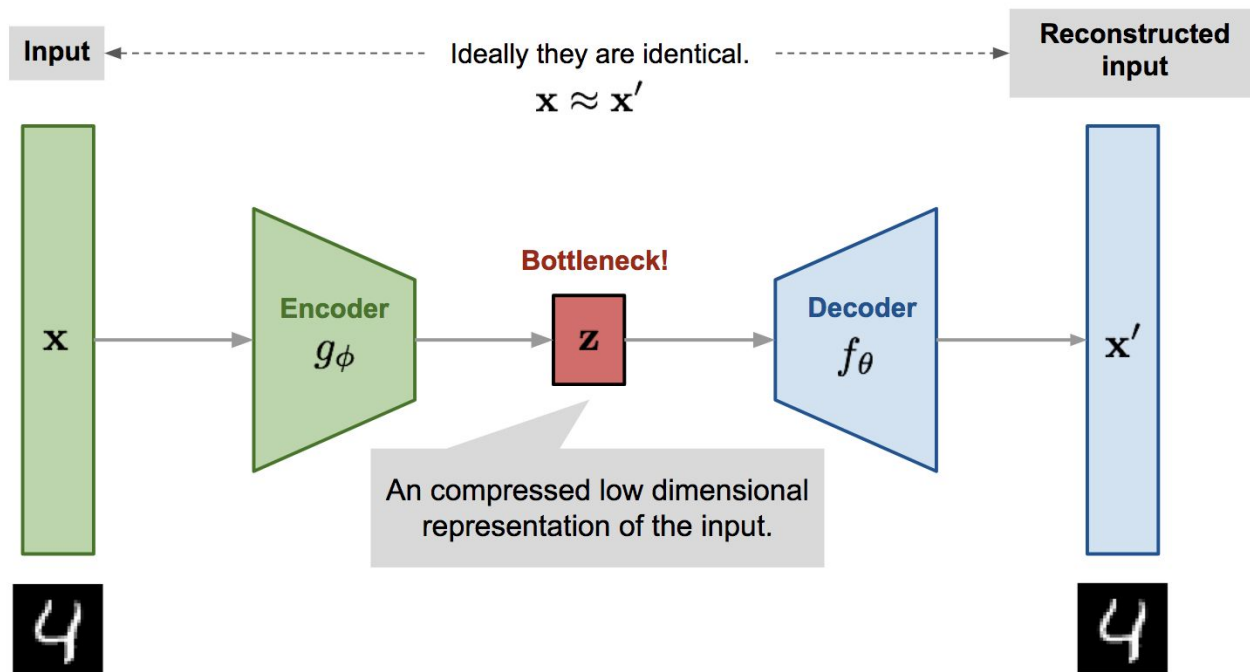
Stable Diffusion은 다음 모델들의 결합

- U-Net
- VAE
- CLIP(Text Encoder / Image Encoder)

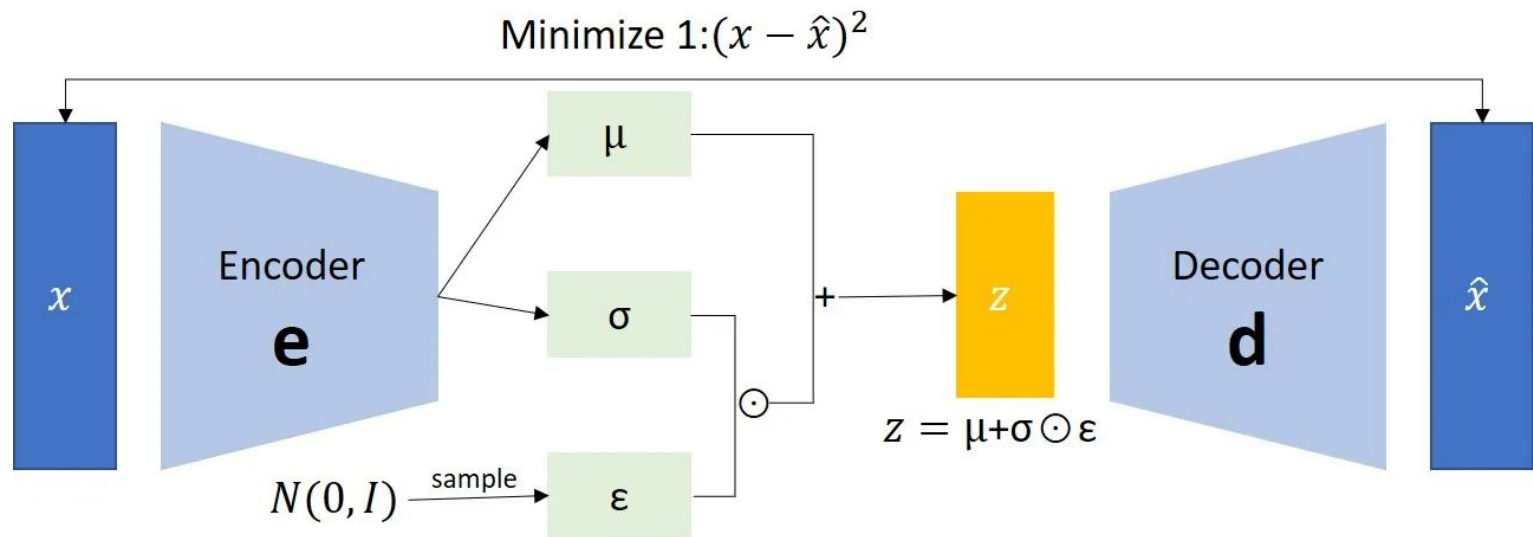
U-Net



Autoencoder



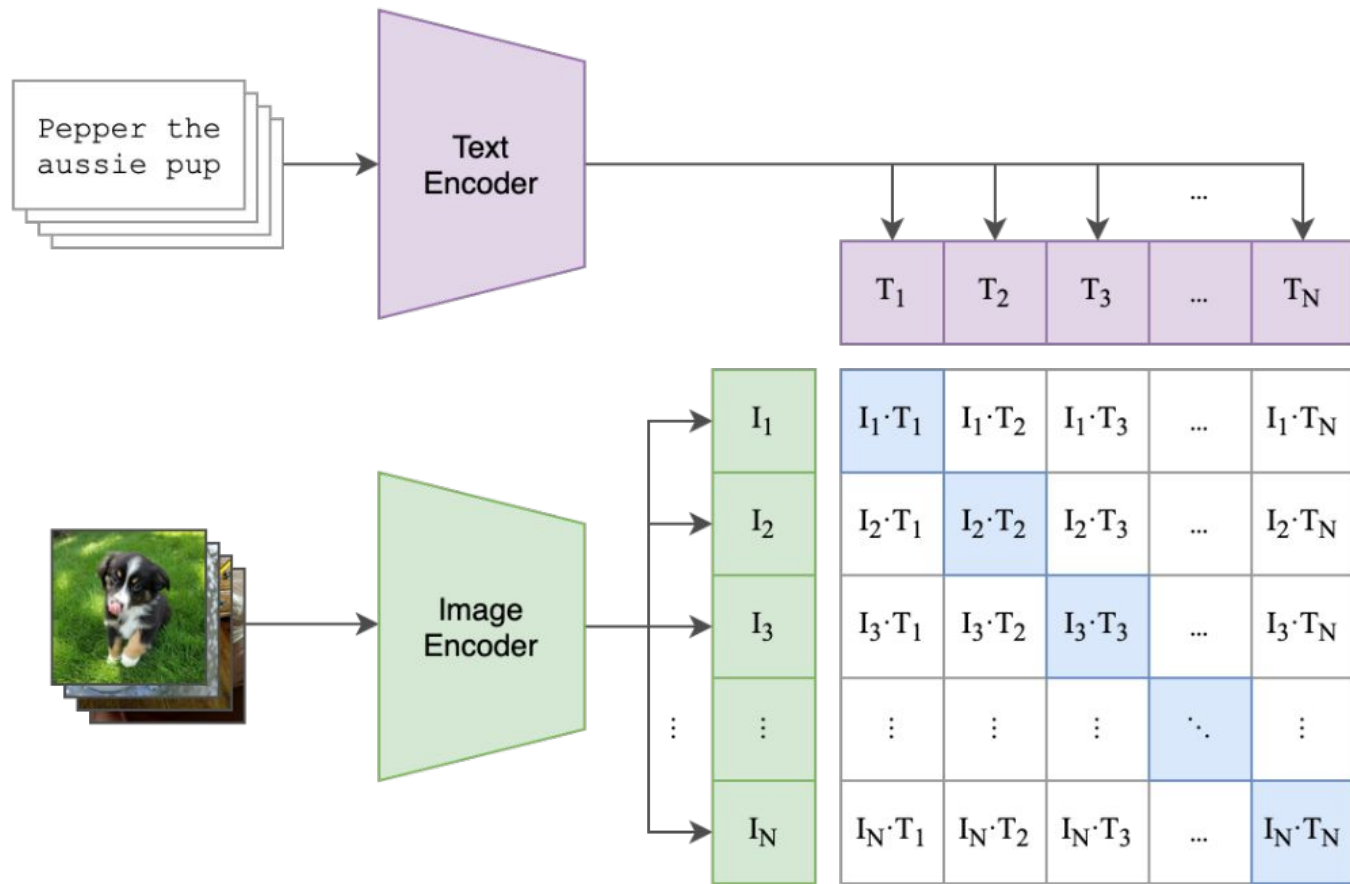
VAE



Minimize 2: $\frac{1}{2} \sum_{i=1}^N (\exp(\sigma_i) - (1 + \sigma_i) + \mu_i^2)$

CLIP

(1) Contrastive pre-training





InfoNCE Loss

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{N} \sum_{i=1}^N \left[\log \frac{\exp(\text{sim}(t_i, v_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(t_i, v_j)/\tau)} \right]$$

Diffusion의 기본 컨셉

noising의 반대
denoising

